

Collision Lab Report

Vuk Tadic

February 24, 2026

1 Purpose

To investigate changes in momentum and kinetic energy during elastic and inelastic collisions.

2 Ground Rules

- Momentum, p , is conserved in isolated systems.
- Kinetic energy, K , is conserved in elastic collisions.
- Kinetic energy K , is not conserved in inelastic collisions.
- $p = mv$
- $K = \frac{1}{2}mv^2$

3 Data

3.1 Elastic Collisions

Table 1: Elastic Collision Data

Collision	Mass 1 (kg)	Mass 2 (kg)	Initial Vel 1 (m s ⁻¹)		Initial Vel 2 (m s ⁻¹)		Final Vel 1 (m s ⁻¹)		Final Vel 2 (m s ⁻¹)	
			x	y	x	y	x	y	x	y
Elastic 1	0.50	1.50	3.00	0.00	0.00	0.00	-1.50	0.00	1.50	0.00
Elastic 2	1.00	1.00	0.76	0.70	-0.74	0.07	-0.62	-0.13	0.64	0.90
Elastic 3	0.10	3.00	0.33	0.00	-3.00	0.01	-6.11	0.12	-2.79	0.01

3.2 Inelastic Collisions

Table 2: Inelastic Collision Data

Collision	Mass 1 (kg)	Mass 2 (kg)	Initial Vel 1 (m s ⁻¹)		Initial Vel 2 (m s ⁻¹)		Final Vel 1 (m s ⁻¹)		Final Vel 2 (m s ⁻¹)	
			x	y	x	y	x	y	x	y
Inelastic 1	0.50	1.50	2.84	0.00	0.00	0.00	0.61	0.06	0.74	-0.02
Inelastic 2	1.00	1.00	1.61	-0.17	-0.85	-0.50	0.62	-0.81	0.14	0.14
Inelastic 3	0.10	3.00	0.85	0.01	-3.00	0.01	-3.05	-0.17	-2.87	0.02

3.3 Images

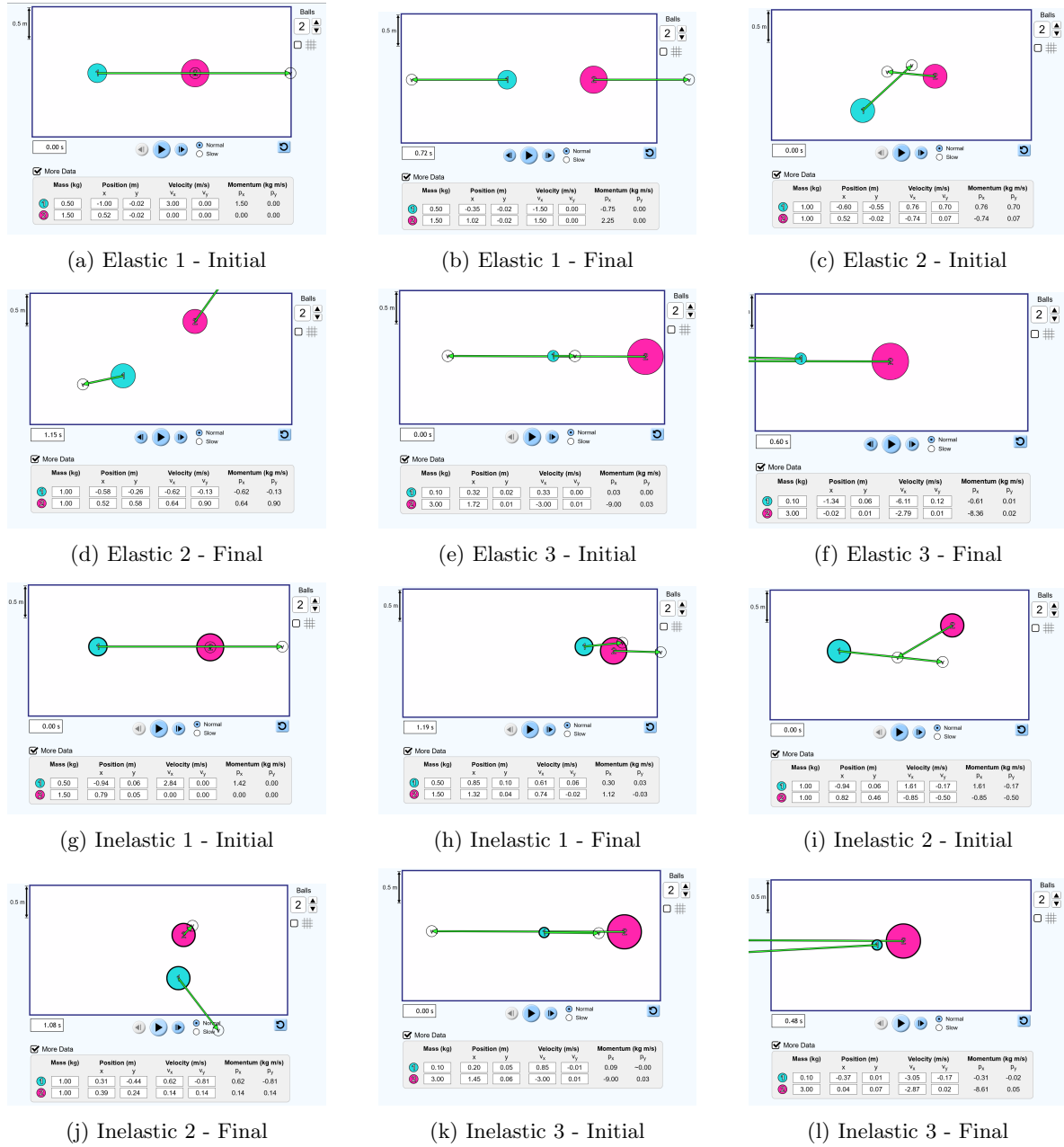


Figure 1: Collision Images

4 Analysis

4.1 Method

For each trial and each object:

$$p = mvK = \frac{1}{2}mv^2$$

System totals are vector sum of momenta and scalar sum of kinetic energies. We compare initial and final totals and report Δ (difference) and a conservation verdict (momentum: magnitude tolerance 0.02 Ns; K: percent threshold 3%).

4.2 Initial Momenta

Table 3: Initial momenta (N s)

Collision	$\mathbf{p}_{1,i,x}$	$\mathbf{p}_{1,i,y}$	$\mathbf{p}_{1,i}$	$\mathbf{p}_{2,i,x}$	$\mathbf{p}_{2,i,y}$	$\mathbf{p}_{2,i}$	$\mathbf{p}_{sys,i,x}$	$\mathbf{p}_{sys,i,y}$	$\mathbf{p}_{sys,i}$
Elastic 1	1.500	0.000	1.500	0.000	0.000	0.000	1.500	0.000	1.500
Elastic 2	0.760	0.700	1.034	-0.740	0.070	0.743	0.020	0.770	0.770
Elastic 3	0.033	0.000	0.033	-9.000	0.030	9.000	-8.967	0.030	8.967
Inelastic 1	1.420	0.000	1.420	0.000	0.000	0.000	1.420	0.000	1.420
Inelastic 2	1.610	-0.170	1.619	-0.850	-0.500	0.986	0.760	-0.670	1.013
Inelastic 3	0.085	0.001	0.085	-9.000	0.030	9.000	-8.915	0.031	8.915

4.3 Final Momenta

Table 4: Final momenta (N s)

Collision	$\mathbf{p}_{1,f,x}$	$\mathbf{p}_{1,f,y}$	$\mathbf{p}_{1,f}$	$\mathbf{p}_{2,f,x}$	$\mathbf{p}_{2,f,y}$	$\mathbf{p}_{2,f}$	$\mathbf{p}_{sys,f,x}$	$\mathbf{p}_{sys,f,y}$	$\mathbf{p}_{sys,f}$	Δp	Conserved
Elastic 1	-0.750	0.000	0.750	2.250	0.000	2.250	1.500	0.000	1.500	0.000	Yes
Elastic 2	-0.620	-0.130	0.633	0.640	0.900	1.104	0.020	0.770	0.770	0.000	Yes
Elastic 3	-0.611	0.012	0.611	-8.370	0.030	8.370	-8.981	0.042	8.981	0.018	Yes - rounding error
Inelastic 1	0.305	0.030	0.306	1.110	-0.030	1.110	1.415	0.000	1.415	-0.005	Yes - rounding error
Inelastic 2	0.620	-0.810	1.020	0.140	0.140	0.198	0.760	-0.670	1.013	0.000	Yes
Inelastic 3	-0.305	-0.017	0.305	-8.610	0.060	8.610	-8.915	0.043	8.915	0.000	Yes

4.4 Kinetic Energy

Table 5: Kinetic Energy (J)

Collision	$\mathbf{K}_{1,i}$	$\mathbf{K}_{2,i}$	$\mathbf{K}_{sys,i}$	$\mathbf{K}_{1,f}$	$\mathbf{K}_{2,f}$	$\mathbf{K}_{sys,f}$	ΔK	Conserved
Elastic 1	2.250	0.000	2.250	0.563	1.688	2.250	0.000	Yes
Elastic 2	0.534	0.276	0.810	0.201	0.610	0.810	0.000	Yes
Elastic 3	0.005	13.500	13.506	1.867	11.676	13.544	0.038	Yes - rounding error
Inelastic 1	2.016	0.000	2.016	0.094	0.411	0.505	-1.511	No
Inelastic 2	1.311	0.486	1.797	0.520	0.020	0.540	-1.257	No
Inelastic 3	0.036	13.500	13.536	0.467	12.356	12.823	-0.714	No

Note that there are rounding errors due to the low precision the PhET simulation.

4.5 Example Calculations

For the Inelastic 2 trial, we can calculate all results values from our raw data.

Momenta:

$$m_1 = 1.00 \text{ kg}, \quad m_2 = 1.00 \text{ kg}$$

$$v_{1,i,x} = 1.61 \text{ m s}^{-1}, \quad v_{2,i,x} = -0.85 \text{ m s}^{-1}$$

$$v_{1,i,y} = -0.17 \text{ m s}^{-1}, \quad v_{2,i,y} = -0.50 \text{ m s}^{-1}$$

$$v_{1,f,x} = 0.62 \text{ m s}^{-1}, \quad v_{2,f,x} = 0.14 \text{ m s}^{-1}$$

$$v_{1,f,y} = -0.81 \text{ m s}^{-1}, \quad v_{2,f,y} = 0.14 \text{ m s}^{-1}$$

$$p_{1,i,x} = m_1 v_{1,i,x} = 1.00 \text{ kg} \cdot 1.61 \text{ m s}^{-1} = 1.610 \text{ N s}, \quad p_{2,i,x} = -0.850 \text{ N s}$$

$$p_{1,i,y} = -0.170 \text{ N s}, \quad p_{2,i,y} = -0.500 \text{ N s}$$

$$p_{1,f,x} = 0.620 \text{ N s}, \quad p_{2,f,x} = 0.140 \text{ N s}$$

$$p_{1,f,y} = -0.810 \text{ N s}, \quad p_{2,f,y} = 0.140 \text{ N s}$$

$$|\vec{p}_{1,i}| = \sqrt{p_{1,i,x}^2 + p_{1,i,y}^2} = 1.619 \text{ N s}, \quad |\vec{p}_{2,i}| = 0.986 \text{ N s}$$

$$|\vec{p}_{1,f}| = 1.020 \text{ N s}, \quad |\vec{p}_{2,f}| = 0.198 \text{ N s}$$

$$p_{sys,i,x} = p_{1,i,x} + p_{2,i,x} = 0.760 \text{ N s}, \quad p_{sys,f,x} = 0.760 \text{ N s}$$

$$p_{sys,i,y} = -0.670 \text{ N s}, \quad p_{sys,f,y} = -0.670 \text{ N s}$$

$$|\vec{p}_{sys,i}| = \sqrt{p_{sys,i,x}^2 + p_{sys,i,y}^2} = 1.013 \text{ N s},$$

$$|\vec{p}_{sys,f}| = \sqrt{p_{sys,f,x}^2 + p_{sys,f,y}^2} = 1.013 \text{ N s}$$

$$\Delta p = |\vec{p}_{sys,f}| - |\vec{p}_{sys,i}| = 0.000 \text{ N s}$$

$\therefore$ Momentum is conserved.

Kinetic Energy:

$$K_{1,i} = \frac{1}{2} m_1 v_{1,i}^2 = \frac{1}{2} m_1 (v_{1,i,x}^2 + v_{1,i,y}^2) = 1.310 \text{ J}, \quad K_{2,i} = 0.486 \text{ J}$$

$$K_{1,f} = 0.520 \text{ J}, \quad K_{2,f} = 0.020 \text{ J}$$

$$K_{sys,i} = K_{1,i} + K_{2,i} = 1.797 \text{ J}, \quad K_{sys,f} = 0.540 \text{ J}$$

$$\Delta K = K_{sys,f} - K_{sys,i} = -1.257 \text{ J}$$

$\therefore$ Kinetic energy is not conserved.

5 Explanation

The diagrams below show the total momentum of the system combined with the components of the individual momenta for the Inelastic 2 trial. It shows that the system momentum stays constant despite the components changes.

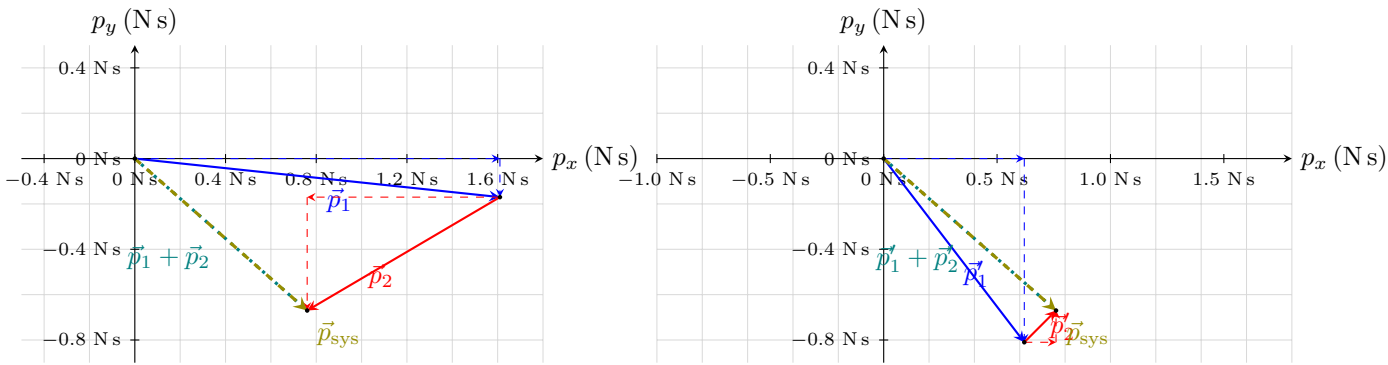


Figure 2: Tip-to-tail addition of components of momenta

Initial (Inelastic 2): $p_{1,i,x} = 1.610 \text{ N s}$, $p_{1,i,y} = -0.170 \text{ N s}$,
 $p_{2,i,x} = -0.850 \text{ N s}$, $p_{2,i,y} = -0.500 \text{ N s}$,
 $p_{\text{sys},x} = p_{1,i,x} + p_{2,i,x} = 0.760 \text{ N s}$, $p_{\text{sys},y} = p_{1,i,y} + p_{2,i,y} = -0.670 \text{ N s}$

Final (Inelastic 2): $p_{1,f,x} = 0.620 \text{ N s}$, $p_{1,f,y} = -0.810 \text{ N s}$,
 $p_{2,f,x} = 0.140 \text{ N s}$, $p_{2,f,y} = 0.140 \text{ N s}$,
 $p_{\text{sys},x} = p_{1,f,x} + p_{2,f,x} = 0.760 \text{ N s}$, $p_{\text{sys},y} = p_{1,f,y} + p_{2,f,y} = -0.670 \text{ N s}$

In both the initial and final states, the system momentum is the same. Despite the x and y components being different, they still add up to the same point in space (0.760 N s right and 0.670 N s down). This can be explained using Newton's third law of motion:

For every action, there is an equal and opposite reaction.

When you push on an object, it pushes back on you with the same amount of force. If you were ice skating with a friend of the same weight and pushed on them, you would move in opposite directions at the same speed. You would have equal, but opposite momenta.

When stationary, your total momentum is zero. After pushing, your total momentum is still zero because they cancel each other out. In a closed system, you can only deliver impulses to each other. The total momentum cannot change because any change in your momentum would be balanced by an equal and opposite change in your friend's momentum.